

Exploration

More on Planning & Control

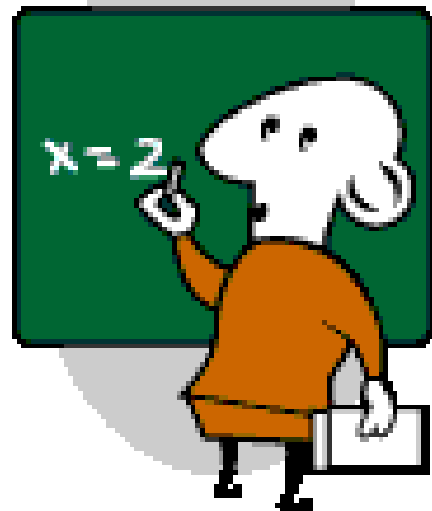
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*Lecture slides heavily use material from the textbook and
Sebastian Thrun, Lecture Slides; <http://www.probablistic-robotics.org/>*



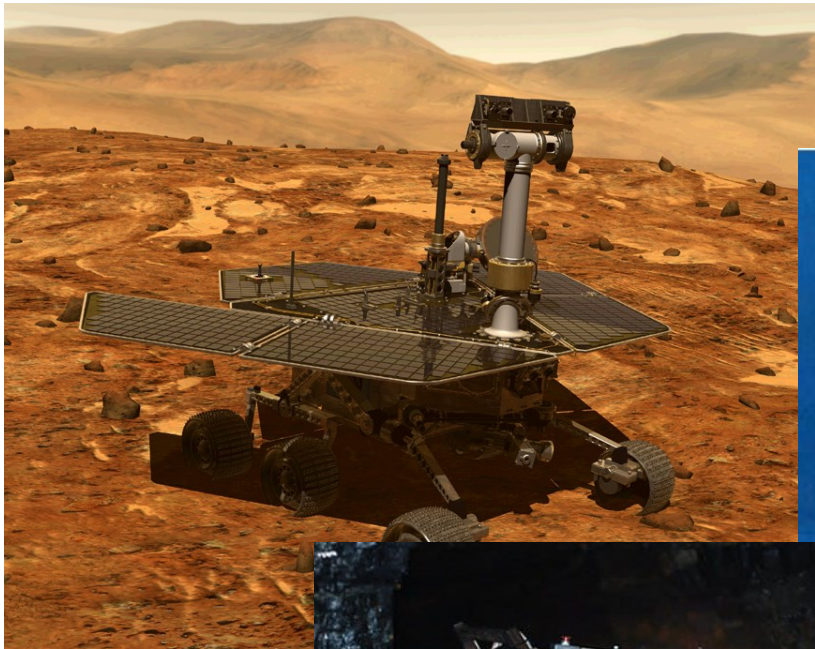
What we will discuss

- Motivations for automated exploration,
- Difficulties with MDPs and POMDPs for exploration problems,
- The basic idea behind successful exploration algorithms: *Maximizing Information Gain*,
- *Monte Carlo Exploration* algorithm,
- Exploration for Active Localization Example,
- Exploration for Occupancy Grid Mapping Example: How to compute Information Gain?
- Extension to Multi-Robot Systems,
- Summary





Motivation for Exploration



Images: http://nssdc.gsfc.nasa.gov/planetary/image/mars2003_rover.jpg
<http://blog.wired.com/defense/images/2008/05/25/titanic6.jpg>
http://www.frc.ri.cmu.edu/robots/robs/photos/2003_Groundhog.jpg



Motivation for Exploration

- **Definition of “Exploration”:**
“Controlling a robot to maximize its (hence our) knowledge about the external world”
- **For humans to explore, environments may be:**
 - *Dangerous,*
 - *Uneconomical,*
 - *Plainly inaccessible (scale)*
- **Robotics may be the only way of collecting information.**
- **Examples:**
 - Planetary exploration, mapping an unknown environment,
 - Searching for/Locating moving entities (people, animals, other robots)
 - Disaster sites (Natural, Nuclear, etc)
 - Underwater, underground, extreme conditions



Motivation for Exploration

- Acquiring information may be a requirement for reliably performing other tasks for the robot.
- Hence:

Effective Exploration is a crucial part of Mobile Intelligent Systems (Robotics) Research



Relations with MDPs and POMDPs

- Markov Decision Processes are mechanisms that enable action selection to maximize payoff,
- Partially Observable MDPs are extensions to noisy sensor case,
- Question:

How can one use MDPs (POMDPs) for exploration?



Relations with MDPs and POMDPs

- MDPs framework covers exploration by defining a suitable payoff function,
- The payoff function should take into account the cost of exploring and the reward of gaining new information,
- These are already discussed...
- What is the difficulty?
- The difficulty is: The number of possible states and possible observations for exploration problems is huge!!
- Makes the value function computation impossible,
- Hence: Other approaches are necessary.



Information Gain - Entropy

- Behind exploration is the concept of “information”,
- By exploring, we seek to gain new information,
- Information Theory gives us tools to measure amount of information carried by random variables:
- Definition: Entropy
“The Entropy of a probability distribution $p(x)$ is the expected information carried by random variable x .”

$$H_p(x) = E[-\log p(x)] = -\int p(x) \log p(x) dx$$

$$H_p(x) = -\sum_x p(x) \log p(x)$$

(unit is “bits” for base 2 log)



Information Gain - Entropy

- Entropy of $p(x)$ maximum when $p(x)$ is a uniform distribution,
- Entropy minimum when $p(x)$ is an impulse function (or point mass function).
- We will use “Conditional Entropy” as a basis for action selection.
- Let $B(b, z, u)$ represent the our conditional belief after executing control u and making observation z at belief state b , then:

$$H_b(x' | z, u) = - \int B(b, z, u)(x') \log B(b, z, u)(x') dx'$$

*Computed using
the Bayes Filter*

*Conditional Entropy after selecting action
 u and observing z starting with b*



Information Gain - Entropy

- We can only pick u and not measurement z , hence measurement is integrated out:

$$\begin{aligned} H_b(x' | u) &\approx E_z[H_b(x' | z, u)] \\ &= \int \int H_b(x' | z, u) p(z | x') p(x' | u, x) b(x) dz dx' dx \end{aligned}$$

- Hence the “information gain” as a result of action u is given by:

$$I_b(u) = H_p(x) - E_z[H_b(x' | u)]$$



Maximizing Information Gain

- How should we choose actions?
- We may seek to maximize “expected information gain”,
- But we also need to take into account the cost of actions (which may have different costs!):

The Payoff Function

- **Greedy Techniques: Maximize the 1-step payoff function:**

$$\pi(b) = \operatorname{argmax}_u \alpha \underbrace{(H_p(x) - E_z[H_b(x' \mid z, u)])}_{\text{expected information gain}} + \underbrace{\int r(x, u) b(x) dx}_{\text{expected costs}}$$



Maximizing Information Gain

- This action selection function becomes:

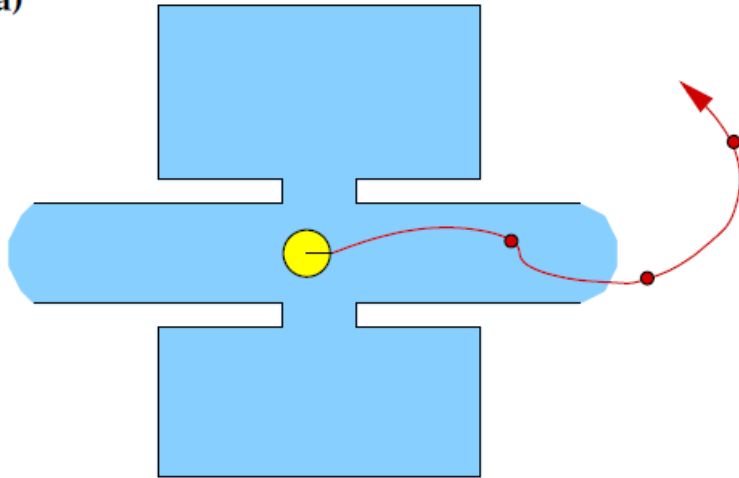
$$\begin{aligned}\pi(b) &= \operatorname{argmax}_u -\alpha E_z[H_b(x' | z, u))] + \int r(x, u) b(x) dx \\ &= \operatorname{argmax}_u \int [r(x, u) - \alpha \int \int H_b(x' | z, u) p(z | x') \\ &\quad p(x' | u, x) dz dx'] b(x) dx\end{aligned}$$



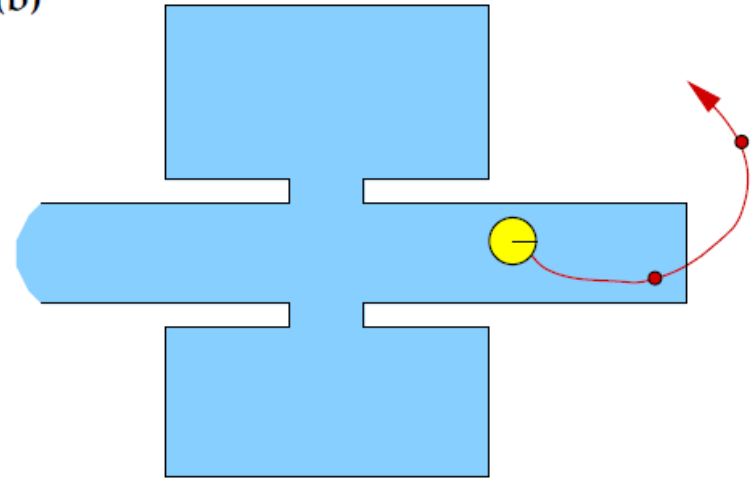
Maximizing Information Gain

- **Question: Why are 1-step Greedy Techniques much more popular for exploration than N-step versions?**

(a)



(b)





Monte Carlo Exploration

- Simple, sampling based exploration algorithm,
- Used as approximation of the action selection rule $\pi(b)$

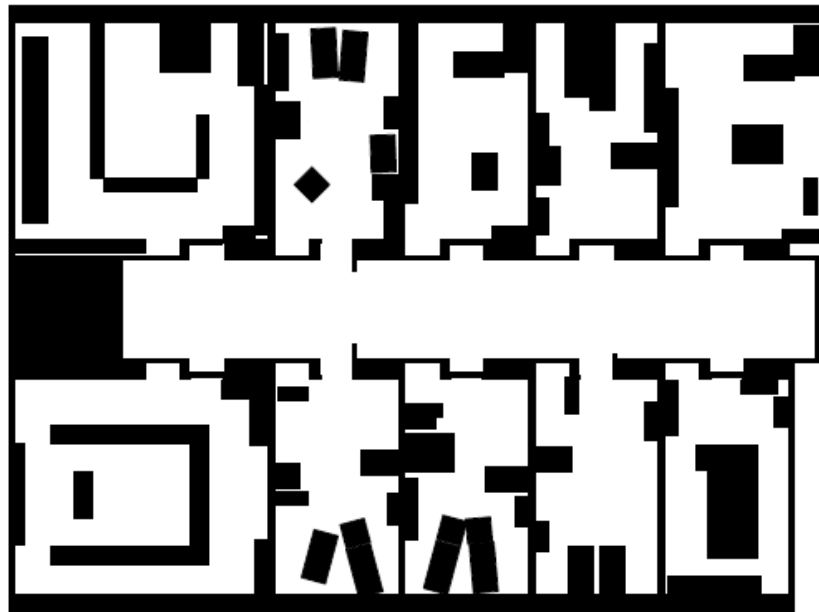
In practice, this simple algorithm may suffer from the curse of dimensionality

```
1:   Algorithm Monte_Carlo_exploration(b):  
2:       set  $\rho_u = 0$  for all actions  $u$   
3:       for  $i = 1$  to  $N$  do  
4:           sample  $x \sim b(x)$   
5:           for all control action  $u$  do  
6:               sample  $x' \sim p(x' | u, x)$   
7:               sample  $z \sim p(z | x')$   
8:                $b' = \text{Bayes\_filter}(b, z, u)$   
9:                $\rho_u = \rho_u + r(x, u) - \alpha H_{b'}(x')$   
10:          endfor  
11:      endfor  
12:      return  $\operatorname{argmax}_u \rho_u$ 
```



Active Localization

- Task is to maximize information about robot pose,
- Much lower dimensional problem with map of environment m given,



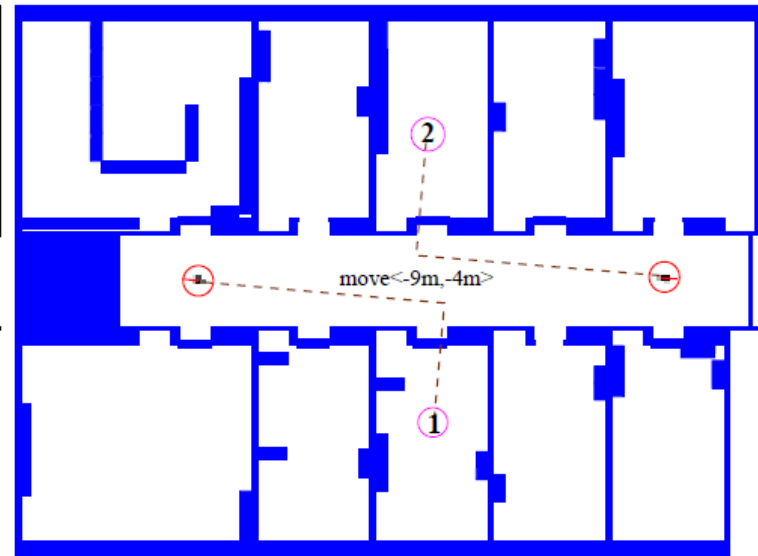
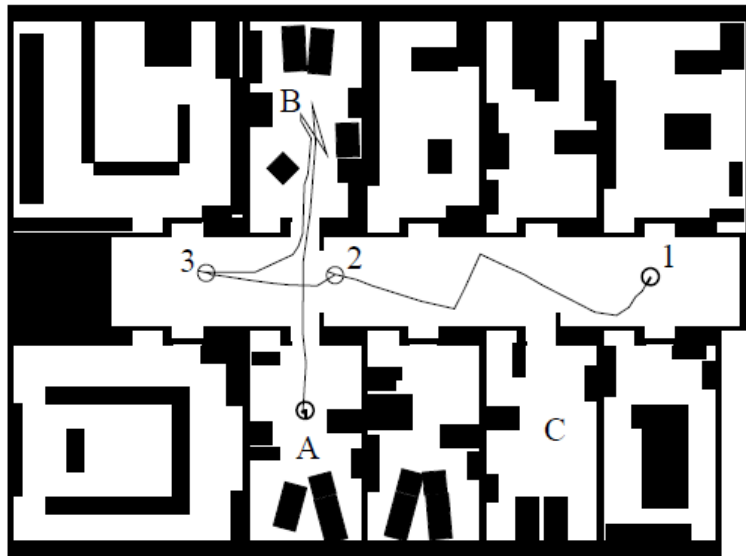
- Exploration algorithms generally very successful for this case!



Active Localization

- We need to define high level “exploration actions” to facilitate problem formulation and solution,
- Exploration actions relative to the robot pose,
- Low level actions are assumed to be generated by a low level “motion planner”,

(a) Environment with an example posterior. (b) Effect of exploration action.

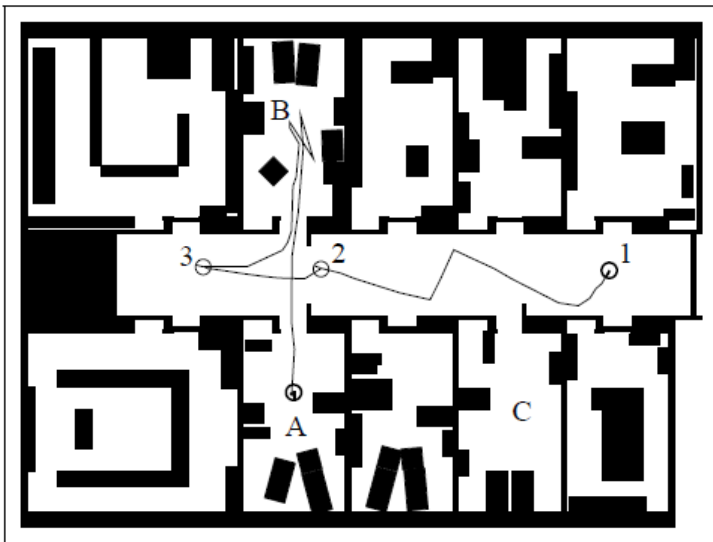




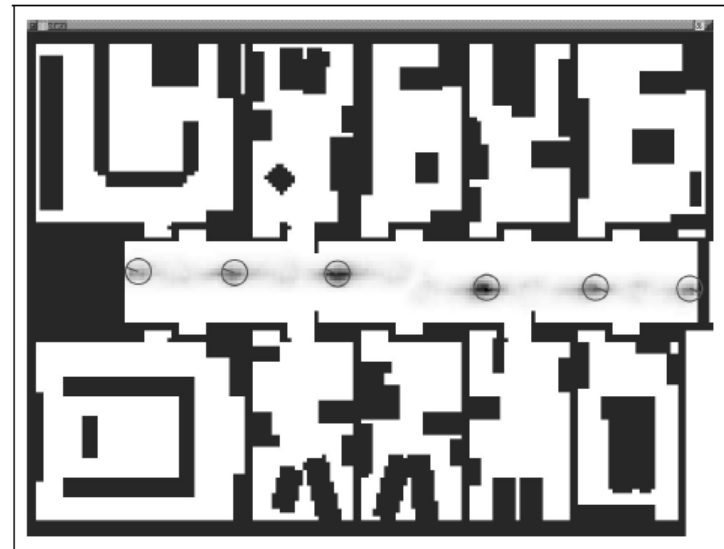
Active Localization: Example (Step 1)

- This formulation enables direct application of the “Monte Carlo Exploration” algorithm:

(a) Path of the localizing robot



(b) Early belief distribution with six modes

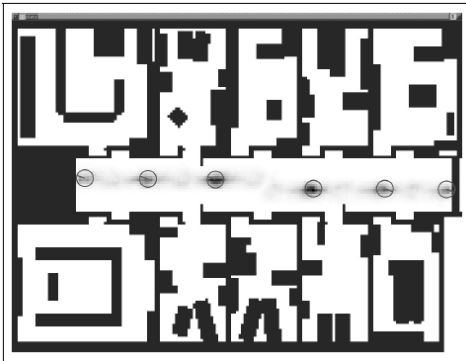


Robot is at location “2”

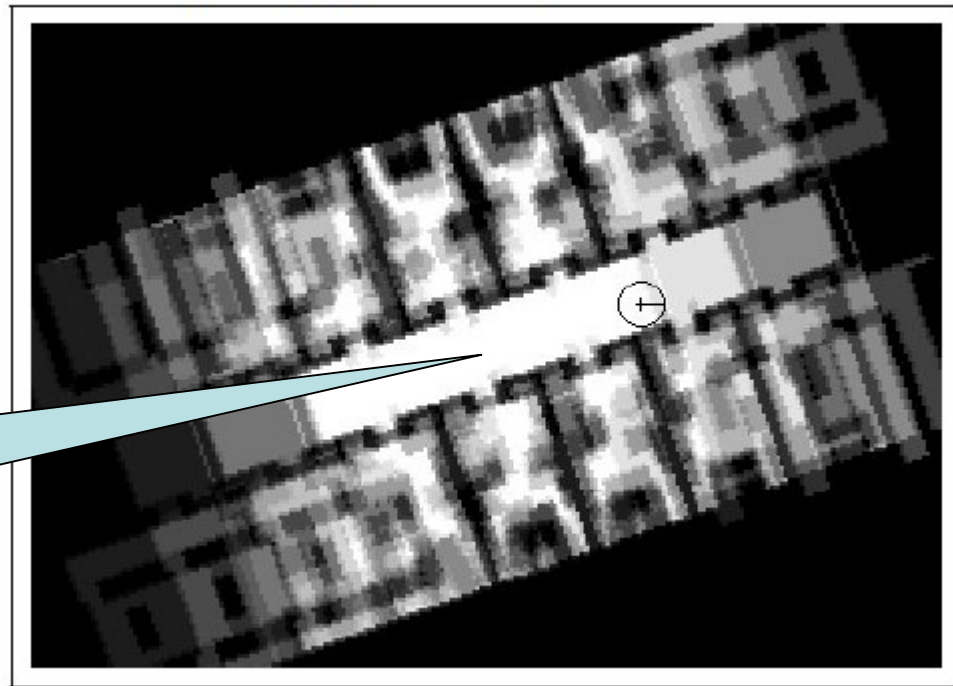


Active Localization: Example (Step 1)

- Occupancy probability computed by conditional averaging over distinct pose hypothesis:



(c) Occupancy probability in robot coordinates



For all likely poses, the robot is in a corridor like traversable area.

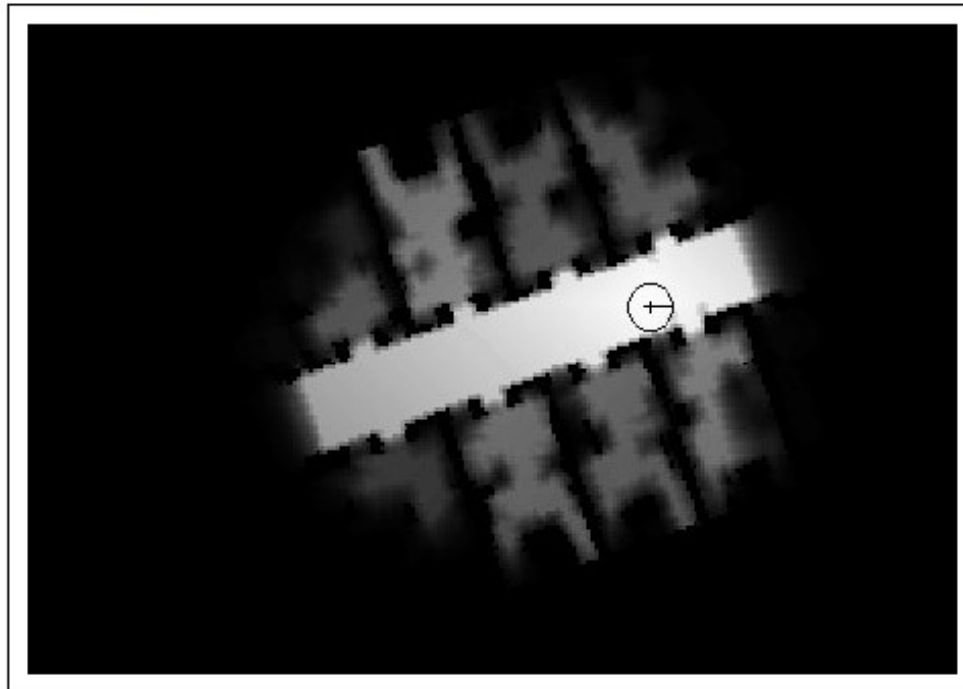
Obtained by superimposing known occupancy grid map for (relative to) each robot pose, weighted by the respective “pose likelihood”.



Active Localization: Example (Step 1)

- Expected cost of motion is computed for all target cells relative to the robot pose using Value Iteration,
- The value function is computed from robot outward,

(d) Expected costs of motion

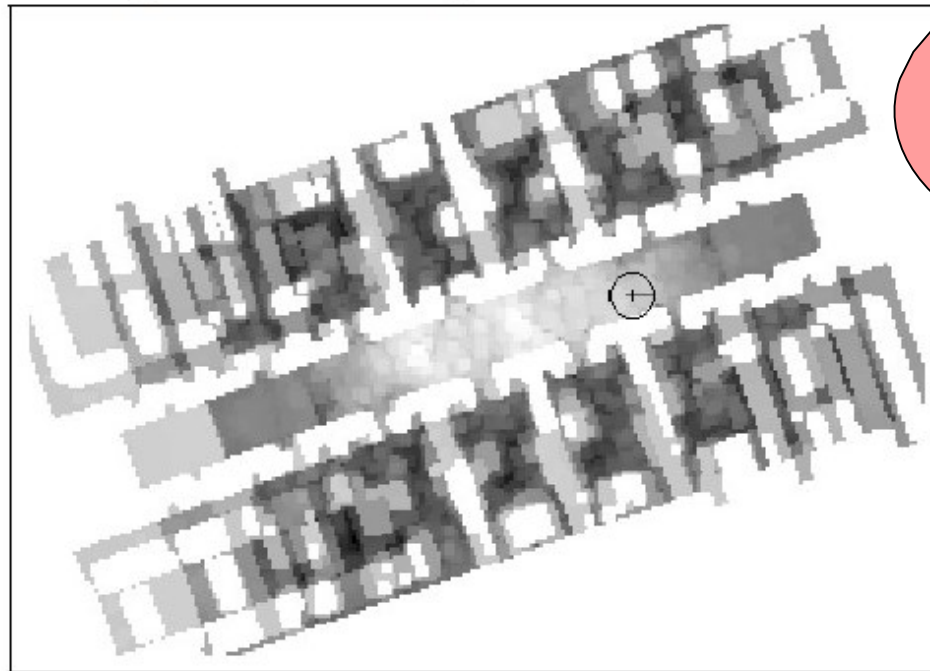




Active Localization: Example (Step 1)

- Expected information gain for any target cell is computed by virtually “placing” the robot at a cell, simulate possible range measurements, integrate, measure information after the Bayes Filter Update. Repeat for every location to get a “map”

(e) Exp. information gain in robot coordinates



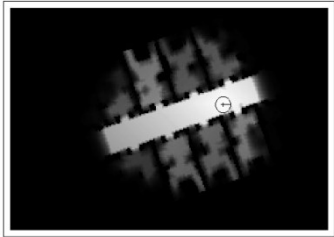
*Darker areas
indicate higher
information
gain.*



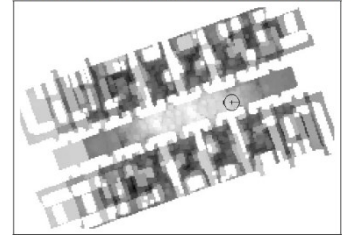
Active Localization: Example (Step 1)

- Expected gain added to the expected costs to determine the action selection function:

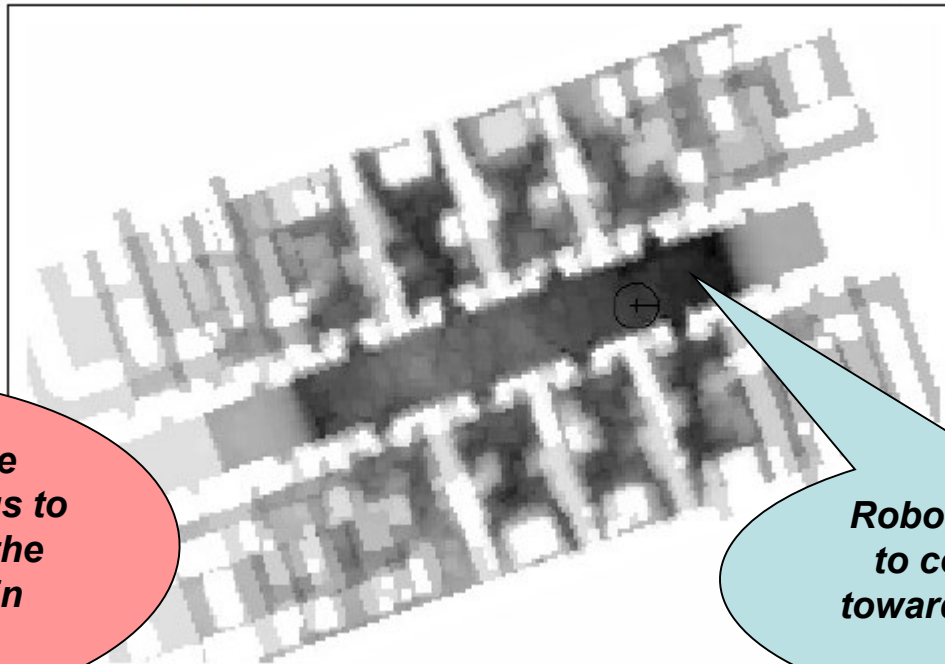
(d) Expected costs of motion



(e) Exp. information gain in robot coordinates



(f) Gain plus costs (the darker, the better)



Clearly, a relative motion that takes us to a room or end of the corridor results in highest gain!

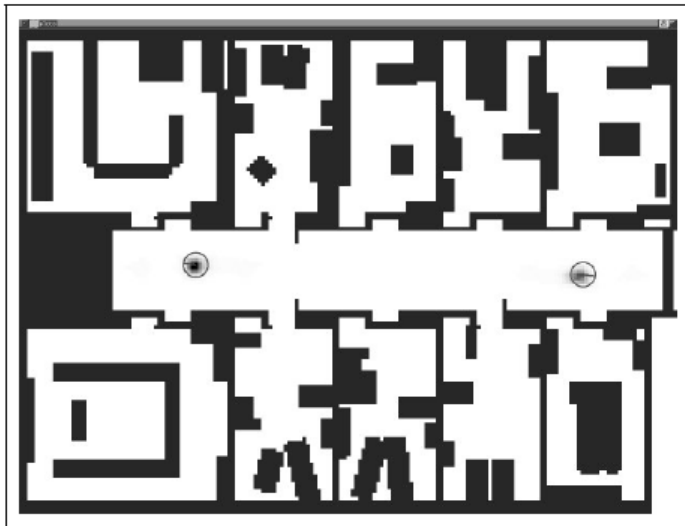
Robot chooses to move to corridor "forward" towards "a corridor end" (location 3)



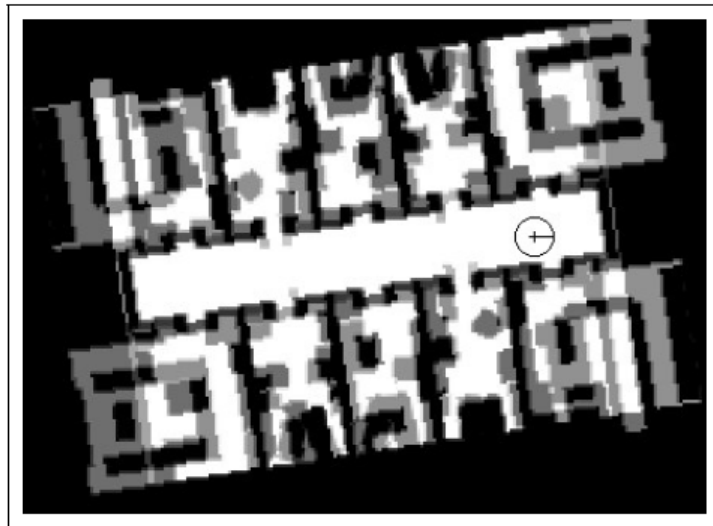
Active Localization: Example (Step 2)

- After motion: New measurements are integrated using the Bayes Filter. Robot pose has two hypothesis now.

(a) Belief distribution



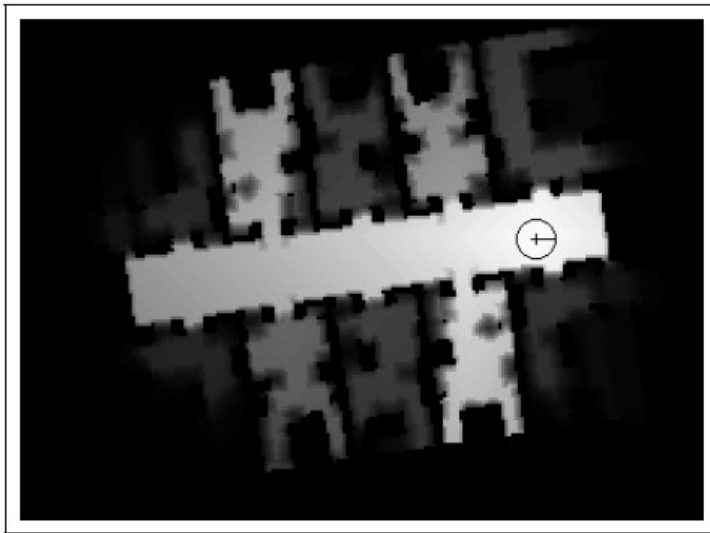
(b) Occupancy prob. in robot coordinates



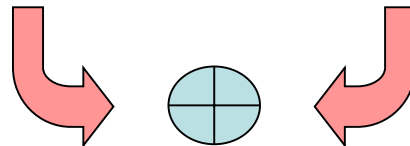
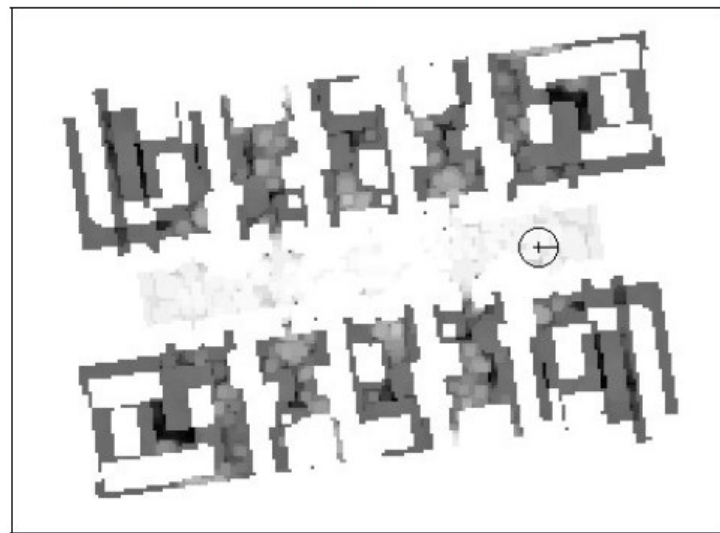


Active Localization: Example (Step 2)

(c) Expected costs of motion



(d) Exp. information gain in robot coordinates

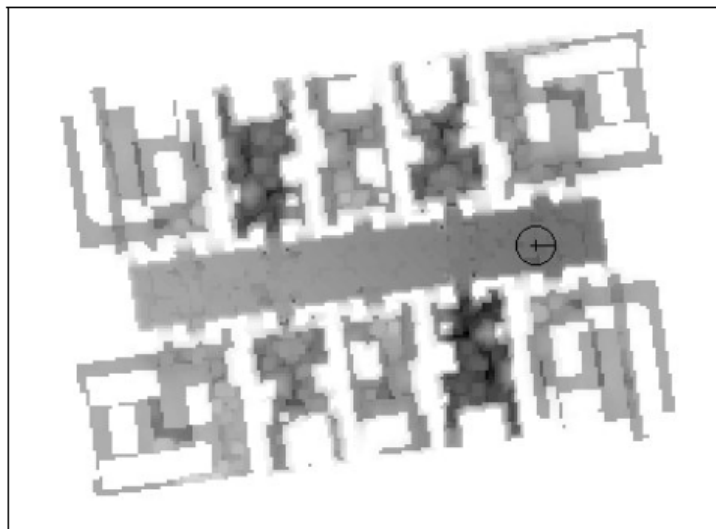




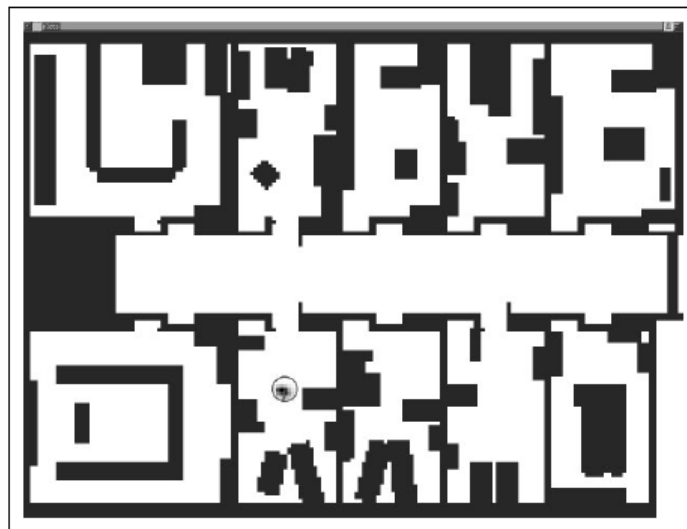
Active Localization: Example (Step 2)

- Robot decides to move “back and left into room”...
- Results in a final localization (Bayes Filter update)

(e) Gain plus costs (the darker, the better)



(f) Final belief after active localization





Exploration for Mapping

- Exploration can be used for mapping, in particular Occupancy Grid Maps.
- We have many more unknown variables, therefore need a further assumption for a feasible application:
- **Assumption: Information gain is independent between each occupancy grid map cells.**
- This is the same assumption that allowed the construction of the occupancy grid mapping algorithm!!



Exploring Occupancy Grid Maps

- Greedy exploration will steer the robot to the nearest possible unexplored area (maximal information gain)
- How to compute the information gain value?
- Three possible techniques to select next exploration target:
 - **Cell Entropy,**
(Simply compute the current entropy of each cell using cell occupancy probability)
 - **Expected Information Gain,**
(Also consider information that would be gained by using the sensors in the vicinity of a map cell)
 - **Binary Gain**
(Simplest and most popular. Just tag all cells that are observed at least once as “explored”, all others as “unexplored”. The gain is a binary function.)
- It turns out that the first two approaches produce very similar action selection functions



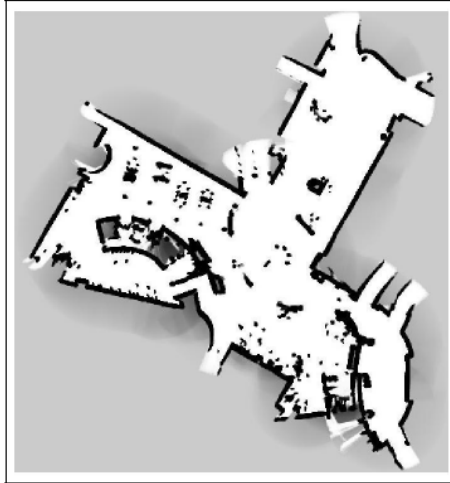
Propagating Gain

- Define an “exploration action” as *“moving to an x-y location along a minimum cost path and sensing the vicinity of the target cell”*
- When we assume a binary gain, it is nonzero only for unexplored boundary cells,
- How can we assign an action selection function for all cells in the map?
- We may use the Value Iteration Algorithm.
- (Boundary cell “gains” becomes a navigation function for all cells in the map)
- Greedy strategy: *Move in $\text{Grad}(V)$ direction!*

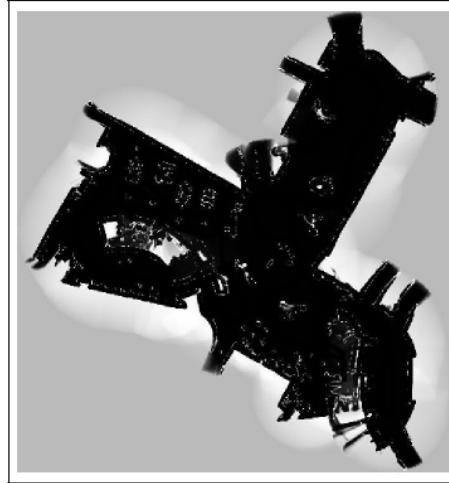


Map Exploration Example

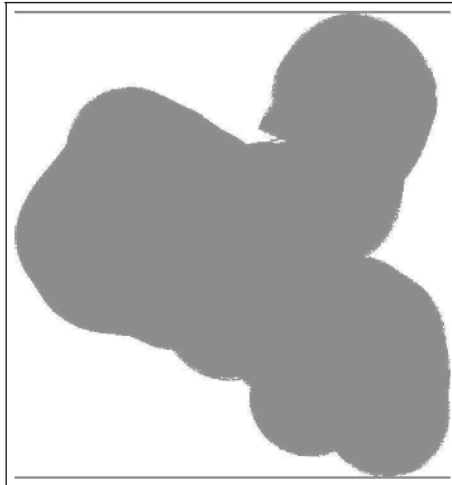
(a) Occupancy grid map



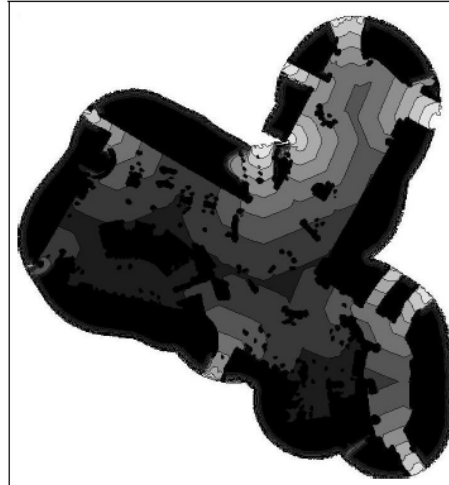
(b) Cell entropy



(c) Explored and unexplored space



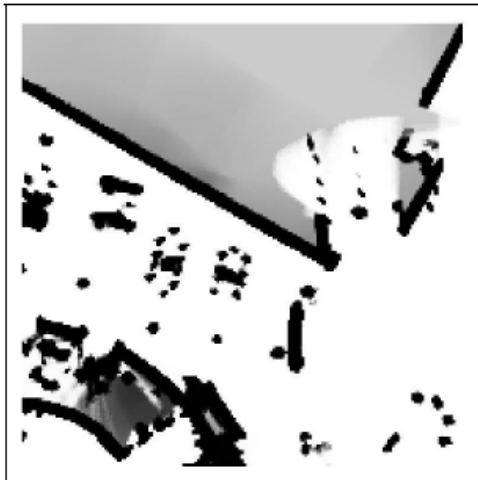
(d) Value function for exploration





Comparison

(a) Map segment



(b) Entropy



(c) Exp. information gain



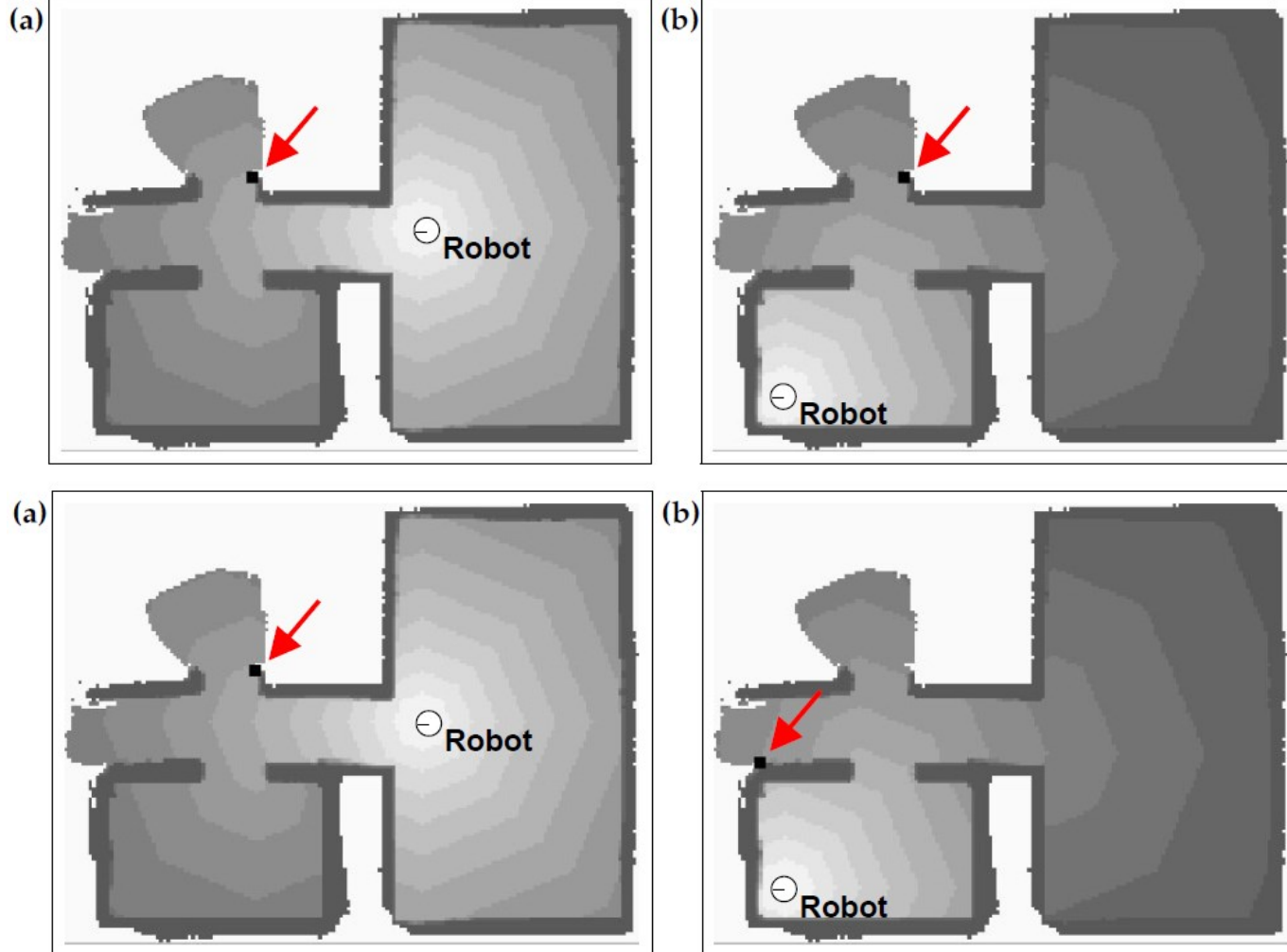


Multi-Robot Exploration

- **The speed-up with the number of robots is generally linear,**
- **The key idea is coordination,**
- **A value function is computed for each robot, propagating outward,**
- **A “task allocation” is done in sequence based on the value function values of each robot.**
- **In a greedy case, each robot picks an exploration goal and prohibits others from picking this goal.**



Multi-Robot Exploration



No coordination

With greedy minimal
coordination



Summary

- **Class's turn to summarize!**